

Iron Sentinel: Network Baseline

Lab Environment and Normal Traffic Patterns

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1. Purpose

This document establishes the normal operating baseline for the Iron Sentinel lab environment. It includes the hosts, open ports, log sources, and traffic patterns observed before any attack simulation began. It exists so that anything abnormal that is observed later in the project (simulated attacks) can be evaluated against a known good state.

2. Hosts in Scope

Host	Role	IP Address	OS
WIN-50QKSHQ5M4N	Target Host	192.168.86.84	Windows Server 2022
ldubyak-VirtualBox	SIEM (Splunk)	192.168.86.89	Ubuntu 22.04 LTS

3. Open Ports and Services

Identified via Nmap Scanning and Windows service enumeration during lab setup

Port	Protocol	Service	Notes
445	TCP	SMB/ microsoft-ds	File sharing. Flagged in vulnerability scan; signing not enforced
3389	TCP	RDP	Remote administration access to Windows Host
135	TCP	MSRPC	Windows RPC endpoint mapper
22	TCP	SSH	Administrative access to the Ubuntu/Splunk Host
8000	TCP	HTTP	Splunk Web interface
8089	TCP	HTTPS	Splunk management port
9997	TCP	Splunk-to-Splunk	Universal Forwarder data channel (Windows - Ubuntu)

4. Log Sources (Splunk)

Three log sources were configured on the Windows host and forwarded to Splunk via the Universal Forwarder over port 9997

Log Source	Sourcetype in Splunk	What it Captures
Windows Security	WinEventLog: Security	Logon/Logoff events, authentication failures, privilege use
Windows System	WinEventLog: System	OS-level service and system events
Sysmon Operational	XMLEventLog: Sysmon	Process creation, network connections, file/registry activity

5. Normal Traffic Observed (Wireshark Capture)

A 5-10 minute packet capture was taken on the Windows host during normal, non-attack activity (browsing, DNS lookups, and Splunk UI access) to establish what routine traffic looks like in this environment.

- DNS queries to resolve standard lookup domains
- ICMP echo request/reply (ping) traffic between the Windows host and the Ubuntu/Splunk host on the network
- HTTP/HTTPS traffic from the Windows host browser to the Splunk web UI (Port 8000) and to general internet endpoints (ex. Tool downloads)
- TCP traffic on port 9997 (forwarded log data from Windows to Splunk)
- No traffic to or from unexpected external hosts, and no DNS queries to suspicious domains

baseline-capture.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
44815	243.366840	192.168.86.84	8.8.8.8	ICMP	74	Echo (ping) request
44824	243.432637	8.8.8.8	192.168.86.84	ICMP	74	Echo (ping) reply
44919	244.430900	192.168.86.84	8.8.8.8	ICMP	74	Echo (ping) request
44927	244.542047	8.8.8.8	192.168.86.84	ICMP	74	Echo (ping) reply
44983	245.401544	8.8.8.8	192.168.86.84	ICMP	74	Echo (ping) request
44987	245.517040	8.8.8.8	192.168.86.84	ICMP	74	Echo (ping) reply
45059	246.512800	192.168.86.84	8.8.8.8	ICMP	74	Echo (ping) request
45065	246.596520	8.8.8.8	192.168.86.84	ICMP	74	Echo (ping) reply

Frame 44815: Packet, 74 bytes on wire (592 bits), 74 bytes captured
> Ethernet II, Src: PCSysntec-55:4d:c3 (08:00:27:55:4d:c3), Dst:
> Internet Protocol Version 4, Src: 192.168.86.84, Dst: 8.8.8.8
> Internet Control Message Protocol

Type Size
GITIGNORE File 1 KB
MD File 4 KB
Microsoft Edge H... 121 KB

Internet Control Message Protocol

Packets: 64191 · Displayed: 8 (0.0%) · Dropped: 0 (0.0%) · Profile: Default

Windows Server 2022 Standard Evaluation
Windows License valid for 90 days
Build 20348.14_release.210507-1500
6:46 PM
6/16/2026

baseline-capture.pcap

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tcp

No.	Time	Source	Destination	Protocol	Length	Info
80	2.525455	192.168.86.84	192.168.86.84	TCP	1514	80 → 51216 [ACK] Seq=50005 Ack=435 Win=0 Len=0
82	2.505096	192.168.86.84	23.215.177.6	TCP	54	51216 → 80 [ACK] Seq=51465 Ack=55 Win=1027 Len=0
83	2.508996	23.215.177.6	192.168.86.84	TCP	1514	80 → 51216 [ACK] Seq=51465 Ack=55 Win=0 Len=0
84	2.608662	192.168.86.84	23.215.177.6	TCP	54	51216 → 80 [ACK] Seq=51465 Ack=55 Win=1027 Len=0
85	2.678535	192.168.86.84	23.215.177.6	TCP	1514	80 → 51216 [ACK] Seq=51465 Ack=55 Win=0 Len=0
86	2.714669	192.168.86.84	23.215.177.6	TCP	54	51216 → 80 [ACK] Seq=51465 Ack=55 Win=1027 Len=0
87	2.744670	23.215.177.6	192.168.86.84	TCP	1514	80 → 51216 [ACK] Seq=51465 Ack=55 Win=0 Len=0
91	2.805790	192.168.86.84	23.215.177.6	TCP	54	51216 → 80 [ACK] Seq=51465 Ack=55 Win=1027 Len=0
92	2.828720	23.215.177.6	192.168.86.84	TCP	1514	80 → 51216 [ACK] Seq=51465 Ack=55 Win=0 Len=0
97	2.807780	192.168.86.84	23.215.177.6	TCP	54	51216 → 80 [ACK] Seq=51465 Ack=55 Win=1027 Len=0
104	2.896480	23.215.177.6	192.168.86.84	TCP	1514	80 → 51216 [ACK] Seq=51465 Ack=55 Win=0 Len=0
112	2.956030	192.168.86.84	23.215.177.6	TCP	54	51216 → 80 [ACK] Seq=51465 Ack=55 Win=1027 Len=0
113	2.965432	23.215.177.6	192.168.86.84	TCP	1514	80 → 51216 [ACK] Seq=51465 Ack=55 Win=0 Len=0

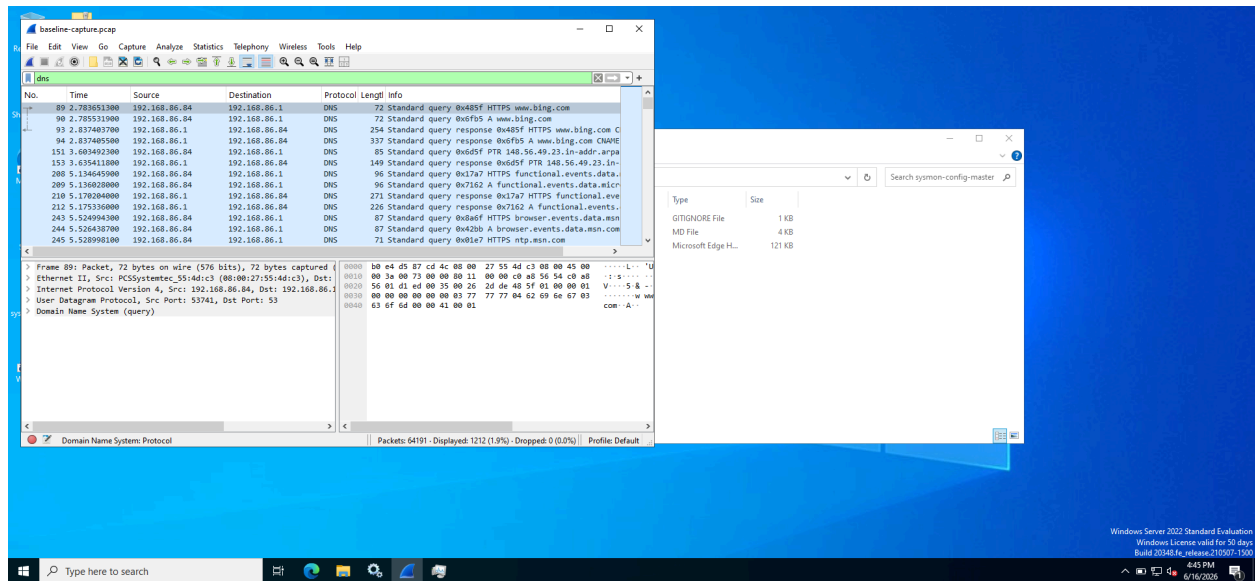
Frame 87: Packet, 1514 bytes on wire (12112 bits), 1514 bytes captured
> Ethernet II, Src: Google_87:cd:4c (08:04:d5:87:cd:4c), Dst: PCSysntec-55:4d:c3 (08:00:27:55:4d:c3)
> Internet Protocol Version 4, Src: 23.215.177.6, Dst: 192.168.86.84
> Transmission Control Protocol, Src Port: 80, Dst Port: 51216, Seq=50005, Win=0, Len=0

Type Size
GITIGNORE File 1 KB
MD File 4 KB
Microsoft Edge H... 121 KB

Transmission Control Protocol

Packets: 64191 · Displayed: 35984 (56.1%) · Dropped: 0 (0.0%) · Profile: Default

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6. Why this matters

Establishing this baseline early made the later detection work meaningful. It is what allowed the brute force, PowerShell obfuscation, and off-hour logon detection rules to be written with a clear sense of what normal looks like on the network instead of guessing on a threshold level. The 720 historical logon events reviewed for the off-hour logon detection rules were judged against the same baseline. All fell within the window established as routine.